

Skills Summary

Handling Ambiguous SSA Cases Completely

Use this reference while completing your worksheet or when studying.

SSA means you know A , its opposite side a , and the adjacent side b . The side a swings from the far vertex like a hinge, so it may meet the base line twice, once, or never. Test before you solve, using the height $h = b \sin A$:

- $a < h$: **no** triangle — the side cannot reach.
- $a = h$: **one**, with a right angle at B .
- $h < a < b$: **two** — the ambiguous case.
- $a \geq b$: **one** — B must be acute, since the larger side faces the larger angle.

The law of sines returns only $\arcsin$, which is always acute. The supplement $180^\circ - B$ has the same sine, and it is a second triangle exactly when $A + (180^\circ - B) < 180^\circ$.

1. Test the case first

Compute $h = b \sin A$, then compare h , a and b before touching the law of sines. One multiplication decides how many answers the problem has — and therefore how many you must report.

Example: $A = 25^\circ$, $b = 18$, $a = 10$. How many triangles?

Step 1. The count depends only on how the swinging side a compares with the height, so compute h before solving anything.

Step 2. $h = 18 \sin 25^\circ$, and since this is less than $a = 10$, which is in turn less than $b = 18$, the measurement gives two triangles.

$$h = 7.61$$

Try it: $A = 35^\circ$, $b = 20$, $a = 8$. Find h and say how many triangles exist.

check: $h = 11.47$, and $a > h$, so none

2. Two triangles: finish both

Find the acute B_1 from $\sin B = \frac{b \sin A}{a}$, then take $B_2 = 180^\circ - B_1$ and confirm $A + B_2 < 180^\circ$. Each B has its own third angle $C = 180^\circ - A - B$ and its own side $c = \frac{a \sin C}{\sin A}$. Report both triangles.

Example: $A = 32^\circ$, $b = 15$, $a = 10$. Find both values of B .

Step 1. Here $h = 7.95 < a < b$, so two triangles exist and the acute answer alone would be half the solution.

Step 2. $\sin B = \frac{15 \sin 32^\circ}{10} = 0.7949$ gives the acute value below, and its supplement 127.36° passes the test since $32^\circ + 127.36^\circ < 180^\circ$.

$$B_1 = 52.64^\circ$$

Try it: $A = 41^\circ$, $b = 22$, $a = 16$. Find the acute B , then its supplement.

check: $B_1 = 64.43^\circ$, so $B_2 = 115.57^\circ$

3. One triangle: show the other candidate fails

When $a \geq b$ the supplement always fails, because $A + (180^\circ - B)$ comes out above 180° . Say so explicitly — “ B_2 rejected, $A + B_2 > 180^\circ$ ” is the line that shows you tested rather than forgot.

Example: $A = 55^\circ$, $b = 9$, $a = 14$. Find B and justify that it is the only one.

Step 1. Since $a = 14$ is longer than $b = 9$, the angle opposite a is the larger one, so B cannot be obtuse and one triangle is guaranteed.

Step 2. $\sin B = \frac{9 \sin 55^\circ}{14} = 0.5266$, and the supplement 148.22° would give $A + B_2 = 203.22^\circ > 180^\circ$, so it is rejected.

$$B = 31.78^\circ$$

Try it: $A = 63^\circ$, $b = 12$, $a = 20$. Find B and say in one line why there is no second triangle.

check: $B = 32.32^\circ$, and $a > b$ rejects the supplement